

Executive Summary

Capstone Project – Real World Business Analysis

This capstone project presents an end-to-end business analysis using real-world datasets to demonstrate a complete data analytics workflow from data preparation to business recommendations. The objective was to apply data-driven methods to identify patterns, understand business performance, and generate actionable insights that support strategic decision-making.

The analysis combines multiple datasets related to sales performance, customer churn behavior, and house pricing trends. The project follows a structured approach beginning with data cleaning and preparation, followed by exploratory data analysis (EDA), advanced analytical techniques, and final insight-driven decision support. This workflow reflects industry-standard analytics practices and emphasizes reproducibility and clear documentation.

During the data preparation phase, raw data was cleaned by handling missing values, removing duplicates, and preparing features for analysis. Exploratory analysis was then performed to examine distributions, detect outliers, and identify relationships between variables. Visualization techniques such as trend charts, correlation heatmaps, and comparative analysis were used to uncover meaningful patterns within the data. Advanced analysis focused on evaluating business key performance indicators (KPIs), understanding feature relationships, and extracting insights relevant to business strategy.

Key findings from the analysis indicate that pricing-related variables have a strong relationship with overall sales performance, suggesting opportunities for optimization through pricing strategies. Customer churn analysis revealed identifiable segments that may require targeted retention efforts. Additionally, trend and correlation analysis highlighted factors that significantly influence business outcomes, supporting data-driven decision-making.

Based on the findings, several recommendations were proposed. These include optimizing pricing strategies using analytical insights, implementing targeted retention initiatives for high-risk customer groups, increasing focus on high-performing business segments, and monitoring key metrics through data dashboards for continuous improvement. A phased implementation approach is recommended, beginning with KPI monitoring and strategy deployment, followed by evaluation and refinement based on measured results.

Overall, this project demonstrates the ability to perform a complete analytics lifecycle by combining technical analysis with business reasoning. The results highlight how structured data analysis can convert raw datasets into meaningful insights that support strategic planning and organizational growth.